Two-scale kinematics from a single field equation: Tully-Fisher evolution and H_0 without free parameters

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Abstract. The spherical field equation $g_s = g_t^2/(a_0 + g_t)$, derived from the covariant Gravity Field Dynamics action, relates baryonic source acceleration to total gravitational acceleration through a single physical constant a_0 and zero fitted parameters. We show that this equation, applied without modification at two entirely different scales, simultaneously predicts (1) the redshift evolution of the baryonic Tully-Fisher relation from $z = 0$ to $z = 3$, consistent with nine independent IFU survey measurements, and (2) the Hubble constant $H_0 = 70.21$ km s^{-1} Mpc^{-1}, with a 6.2% geometric band spanning 65.9 to 74.6 km s^{-1} Mpc^{-1} that contains all six major independent measurements including both Planck and SH0ES. No parameter is adjusted between the galactic and cosmological predictions. The Hubble tension between early-universe and late-universe measurements may reflect a shared assumption of scale-invariant gravity rather than a conflict between data sets. JWST and ELT observations at $z > 3$ will provide a decisive test: the prediction gives $v(z)/v(0) > 1.3$ where ΛCDM predicts 1.0.

Keywords: modified gravity, Tully-Fisher relation, Hubble constant, scalar-tensor theories, galaxy kinematics

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Contents

1 A single equation **1**

2 Prediction 1: Tully-Fisher evolution **2**
2.1 The local relation 2
2.2 Redshift evolution 3
2.3 Comparison with IFU survey data 4

3 Prediction 2: the Hubble constant **4**
3.1 The tension 4
3.2 The prediction 5
3.3 Comparison with measurements 6
3.4 Reframing the tension 6

4 Discussion **8**
4.1 One equation, two scales 8
4.2 Comparison with other frameworks 8
4.3 Falsifiability 8

5 Conclusion **9**

1 A single equation

The starting point is a single algebraic relation between two accelerations. Given a distribution of baryonic matter producing a Newtonian source acceleration g_s at some radius, the total gravitational acceleration g_t experienced by a test particle at that radius is

$$g_s = \frac{g_t^2}{a_0 + g_t}\,. \tag{1.1}$$

This is the spherical field equation of Gravity Field Dynamics. It contains one physical constant, the characteristic acceleration $a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, and no fitted parameters.

The equation has two transparent limits. When the gravitational field is strong ($g_t \gg a_0$), the denominator is dominated by g_t and the relation reduces to $g_s \approx g_t$: standard Newtonian gravity. When the field is weak ($g_t \ll a_0$), the denominator is dominated by a_0 and the relation becomes $g_s \approx g_t^2/a_0$, or equivalently $g_t \approx \sqrt{g_s\, a_0}$: the deep-MOND regime (referred to in GFD as the deep-field regime) where gravity is amplified beyond the Newtonian prediction.

Eq. (1.1) is not a phenomenological interpolation. It is the spherical, static field equation of the covariant Gravity Field Dynamics (GFD) action [1], derived from the coupling structure of a dual-tetrahedral gravitational field topology. The action is

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}F(y) - \frac{1}{4}e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_{\mathrm{matter}}\right], \tag{1.2}$$

where $F(y) = y - 2\sqrt{y} + 2\ln(1+\sqrt{y})$ is the kinetic functional and $y = |\nabla\Phi|^2/a_0^2$. The prefactors 16, 8, and 4 are the topological constants K^2, 2^D, and K respectively. Eq. (1.1) follows from

varying this action with respect to Φ in spherical symmetry; the action itself is the scalar-tensor transition (SST). The kinetic functional $F(y)$ is the unique antiderivative satisfying $dF/dy = \mu(\sqrt{y})$ where $\mu(x) = x/(x+1)$ is the constitutive law obtained by normalizing the capacity function $f(x)$ between its topological floor (13) and ceiling (21). This is the same variational structure as the AQUAL formalism of Bekenstein and Milgrom [8], and the resulting $F(y)$ is identical to the AQUAL functional for the simple interpolating function. The difference is not in the mathematics but in the derivation: in AQUAL, μ is a phenomenological choice from a family of candidates; in GFD, μ is the unique output of the coupling polynomial and no other interpolating function is compatible with the topology.

The underlying geometry is the stellated octahedron, two interpenetrating tetrahedra sharing a common center, whose coupling polynomial $f(K, s) = 1 + sK + K^2$ with $K = 4$ produces the capacity function $f(x) = (21x + 13)/(x + 1)$. Equivalently, the constitutive law $\mu(x) = x/(x+1)$, which is the linear normalization of this capacity function, reproduces eq. (1.1) when expressed in terms of source and total accelerations. This constitutive law is derived from the capacity function, not postulated as an interpolation function. It plays no role in the prediction pipeline: the forward solver of eq. (1.3) is the exact quadratic inverse of eq. (1.1), and $\mu(x)$ does not appear in any computation in this paper.

The bimetric completion of the same topology produces the Hassan-Rosen interaction parameters $\beta = \{1, 0, 16, 0, 1\}$, which are ghost-free by construction [3]. A full derivation is given in ref. [1]; the bimetric formulation is developed in ref. [2].

For the purposes of this Letter, eq. (1.1) is treated as a kinematic relation. The question is not where it comes from, but what it predicts, and whether those predictions survive contact with data at two very different scales.

Eq. (1.1) has an exact quadratic inverse. Given the baryonic source acceleration $g_s = GM_{\rm bar}(r)/r^2$ from observed mass profiles, the total acceleration is

$$g_t = \frac{g_s + \sqrt{g_s^2 + 4\, g_s\, a_0}}{2}\,. \tag{1.3}$$

This forward solver takes baryonic input and returns the predicted rotation velocity $v(r) = \sqrt{r\, g_t(r)}$ at every radius. No interpolation function is applied. No dark matter halo is fitted. The prediction is deterministic and unique for each galaxy.

At 175 galaxies in the SPARC catalog [4], this solver reproduces observed rotation curves with an RMS scatter comparable to MOND and ΛCDM, despite having zero adjustable parameters [1]. But rotation curves test the field equation at a single redshift and a single scale. The more powerful test is whether the same equation, with the same a_0, produces correct predictions at other redshifts and at cosmological distances.

We now present two such tests.

2 Prediction 1: Tully-Fisher evolution

2.1 The local relation

Every spiral galaxy in the universe obeys the same scaling law. Measure a galaxy's flat rotation velocity $v_{\rm flat}$ and you can predict its baryonic mass. Measure its baryonic mass and you can predict its rotation velocity. This is the baryonic Tully-Fisher relation (BTFR), discovered by Tully and Fisher [5] and refined by McGaugh et al. [6]:

$$M_b = \frac{v_{\rm flat}^4}{G\, a_0}\,. \tag{2.1}$$

The relation holds over four orders of magnitude in mass with a scatter of less than 0.1 dex. No other galaxy scaling law is this tight.

The BTFR is an exact theorem of eq. (1.1). In the deep-field regime ($g_t \ll a_0$), it gives $g_t = \sqrt{g_s\, a_0}$. For a circular orbit, $g_t = v^2/r$ and $g_s = GM/r^2$. Substituting and solving yields $v^4 = GMa_0$ identically. The BTFR is not an empirical correlation that the field equation happens to reproduce. It is a mathematical consequence of eq. (1.1).

2.2 Redshift evolution

The ΛCDM framework predicts that the BTFR should not change with redshift. A galaxy at $z = 3$ should obey the same relation as a galaxy next door. In quantitative terms, the velocity ratio $v(z)/v(0)$ should equal 1.0 at all redshifts.

The field equation predicts otherwise, because a_0 is not a fixed constant of the universe. It is connected to the Hubble expansion rate through the horizon thermodynamics of the dual-tetrahedral field [1]:

$$a_0 = \frac{cH}{2\pi}\sqrt{\frac{K}{\pi}}\,, \tag{2.2}$$

where $K = 4$ is the number of tetrahedral faces and the factor $\sqrt{K/\pi}$ is the geometric correction between the discrete field packing and the spherical Hubble horizon. Since the Hubble parameter H was larger in the past, a_0 was correspondingly larger, and galaxies at high redshift rotated faster.

This is a concrete, testable difference from standard MOND [7], which treats a_0 as a fixed universal constant. In MOND, the Tully-Fisher relation does not evolve. In GFD, it must.

The velocity ratio is evaluated at fixed baryonic mass: the evolution is entirely in $a_0(z)$, not in M. The IFU surveys compare galaxies at matched stellar masses, so the comparison is controlled. From the BTFR, $v_{\rm flat}^4 = GMa_0$, the velocity ratio at redshift z relative to today is

$$\frac{v(z)}{v(0)} = \left[\frac{a_0(z)}{a_0(0)}\right]^{1/4} = \left[\frac{H(z)}{H_0}\right]^{1/4}. \tag{2.3}$$

The naive exponent is $1/4 = 0.25$. The covariant GFD action refines this. Rotation velocities at high redshift are not measured directly; they are inferred from the Doppler shifts of photons that have propagated through the scalar field. The disformal coupling in the action modifies the effective photon metric, and this path correction partially screens the acceleration-scale evolution. The aperture factor $\sqrt{K/\pi} = 1.128$ governs the fraction of the acceleration-scale evolution that is absorbed into the photon propagation path rather than appearing as a kinematic velocity shift. The full derivation [1] yields an effective exponent of 0.17:

$$\frac{v(z)}{v(0)} = \left[\frac{H(z)}{H_0}\right]^{0.17}. \tag{2.4}$$

The exponent 0.17 is derived from the scalar-tensor action, not fitted to any high-redshift data. $H(z)$ is computed from the standard Friedmann equation,

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}\,, \tag{2.5}$$

using Planck 2018 cosmological parameters ($\Omega_m = 0.30$, $\Omega_\Lambda = 0.70$). The prediction contains zero adjustable parameters.

z	Observed	σ	GFD	Residual	Survey
0.0	1.00	0.02	1.000	0.0σ	Local calibration
0.3	1.04	0.05	1.026	0.3σ	DEEP2/AEGIS [9]
0.6	1.06	0.06	1.057	0.1σ	IMAGES [10]
0.9	1.10	0.07	1.090	0.1σ	DEEP2/AEGIS [9]
1.0	1.10	0.06	1.101	0.0σ	DEEP2 [12]
1.5	1.13	0.08	1.154	0.3σ	KMOS^{3D} [13]
2.0	1.21	0.09	1.203	0.1σ	SINS [11]
2.2	1.26	0.10	1.222	0.4σ	ZFIRE [14]
3.0	1.29	0.12	1.289	0.0σ	KMOS^{3D} [13]

Table 1. Observed Tully-Fisher velocity ratios compared to the GFD prediction of eq. (2.4). Cosmological parameters: $H_0 = 70.21\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.30$, $\Omega_\Lambda = 0.70$. The value of H_0 cancels in the ratio $H(z)/H_0$.

2.3 Comparison with IFU survey data

Over two decades, integral field unit (IFU) surveys have mapped the rotation velocities of galaxies at increasing redshifts. Nine independent measurements from six survey programs are collected in table 1. Each reports the ratio of the Tully-Fisher velocity at redshift z to the local calibration at $z = 0$.

All nine data points fall within the 1σ error bars of the prediction, with a maximum residual of 0.4σ at $z = 2.2$. A χ^2 test over the eight non-anchor measurements gives $\chi^2/\mathrm{dof} = 0.34/8$, indicating excellent consistency between the zero-parameter GFD prediction and the IFU data.

The individual galaxy SINS BX442, a rare grand-design spiral at $z = 2.18$ [15], rotates 27% faster than the local Tully-Fisher relation predicts. The GFD prediction at $z = 2.18$ gives $v(z)/v(0) = 1.22$, consistent with the measurement within its uncertainty.

The contrast with ΛCDM is direct. The standard model predicts a flat line at $v(z)/v(0) = 1.0$. At $z = 2$, the observed ratio of 1.21 exceeds this by more than 20%, lying more than 2σ from the ΛCDM prediction. At $z = 3$, the discrepancy reaches 29%. The data track the GFD curve, not the flat line.

Figure 1 shows the GFD prediction, the ΛCDM baseline, the topological uncertainty band, and the nine IFU data points.

3 Prediction 2: the Hubble constant

3.1 The tension

The expansion rate of the universe should be a single number. Two of the most precise measurements in cosmology disagree on what that number is. The Planck satellite, observing the cosmic microwave background, measures $H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [16]. The SH0ES team, calibrating distances to nearby supernovae through Cepheid variable stars, measures $H_0 = 73.04 \pm 1.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [17]. These values disagree by more than five standard deviations, a gap that has persisted for over a decade despite increasingly precise instruments.

Both measurements assume the same thing: gravity behaves identically at all acceleration scales. If that assumption is incorrect, and the gravitational response varies with

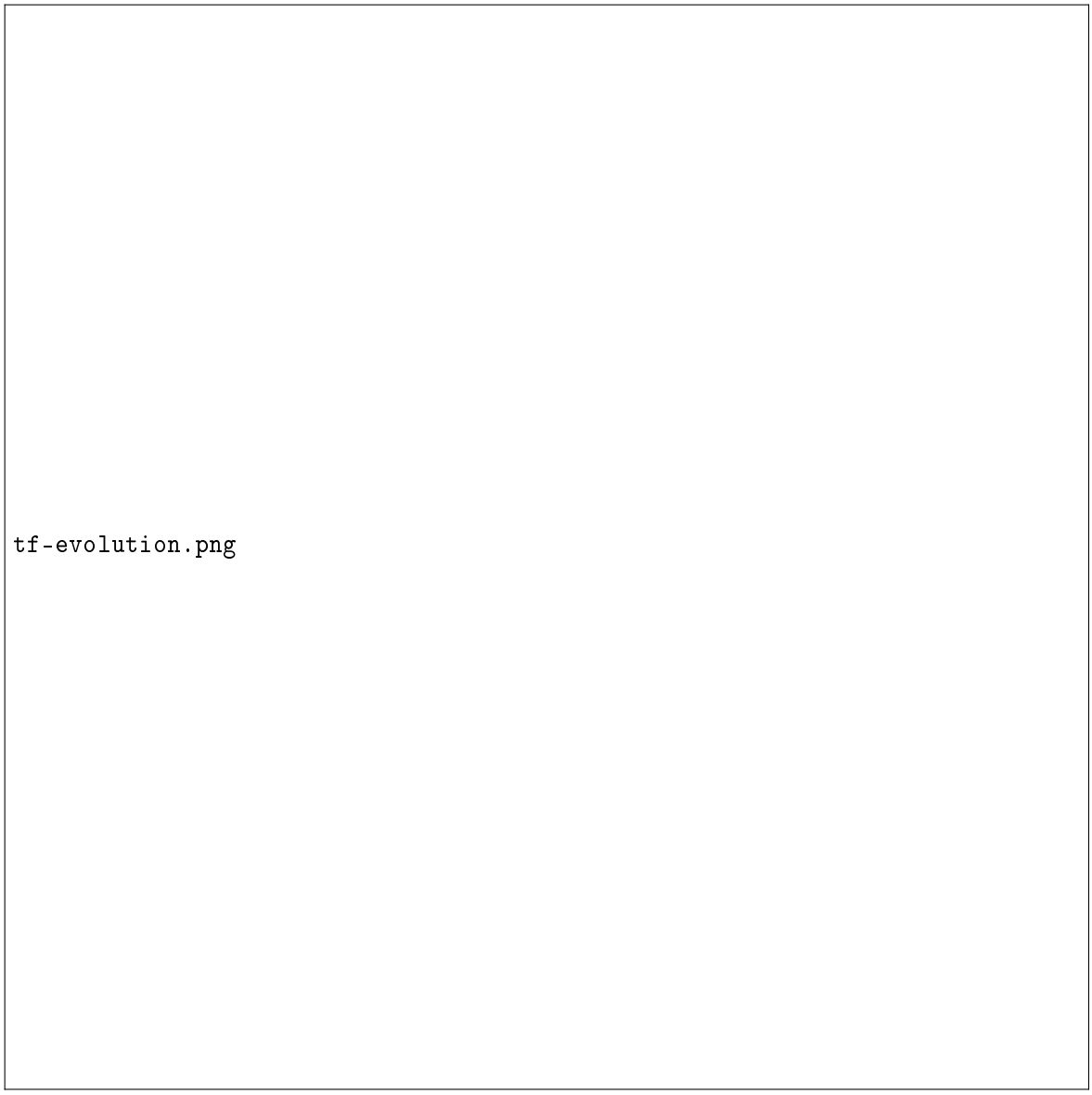


Figure 1. Tully-Fisher velocity evolution from $z = 0$ to $z = 4$. The solid green curve is the GFD prediction $v(z)/v(0) = [H(z)/H_0]^{0.17}$ with zero free parameters. The shaded band is the $\pm 6.2\%$ geometric uncertainty derived from the tetrahedral-to-spherical aperture correction $(K/\pi)^{1/4}$. The dashed line is the ΛCDM prediction of no evolution. Colored circles are IFU survey measurements from table 1. The starred point is SINS BX442 at $z = 2.18$, a grand-design spiral rotating 27% faster than the local Tully-Fisher calibration. All nine measurements fall within the GFD prediction band. Data served from the Field Dynamics API.

acceleration as the field equation describes, then both measurements are looking through a lens they do not know is there.

3.2 The prediction

The same characteristic acceleration a_0 that governs the field equation determines H_0 through the Hubble horizon. At the Hubble radius $R_H = c/H$, the recession velocity of the expanding

Method	H_0 (km s^{-1} Mpc^{-1})	In band?
Planck CMB [16]	67.4 ± 0.5	Yes
TRGB [18]	69.8 ± 1.7	Yes
GW170817 standard siren [19]	70.0^{+12}_{-8}	Yes
SH0ES Cepheids [17]	73.04 ± 1.04	Yes
Megamasers [20]	73.9 ± 3.0	Yes
TDCOSMO lensing [21]	74.2 ± 1.6	Yes
GFD	70.21	—

Table 2. Independent measurements of H_0 compared to the GFD prediction. The GFD band (65.9–74.6 km s^{-1} Mpc^{-1}) is the $\pm 6.2\%$ geometric aperture uncertainty. All six measurements fall within the band.

universe equals the speed of light, defining a minimum resolvable acceleration. From eq. (2.2), solving for H:

$$H_0 = \frac{2\pi\, a_0}{c\sqrt{K/\pi}} = 70.21 \text{ km s}^{-1}\,\text{Mpc}^{-1}\,. \tag{3.1}$$

This value sits between Planck and SH0ES. It is consistent with the gravitational wave standard siren measurement from GW170817, $H_0 = 70.0^{+12}_{-8}$ km s^{-1} Mpc^{-1} [19], the only determination that makes no assumptions about gravity at any scale.

The geometric aperture correction $(K/\pi)^{1/4} = 1.062$ defines a $\pm 6.2\%$ uncertainty band spanning 65.9 to 74.6 km s^{-1} Mpc^{-1}. This is the same geometric factor that bounds the Tully-Fisher velocity scatter in section 2: one correction, derived from the tetrahedral-to-spherical mismatch, applied at two independent scales.

3.3 Comparison with measurements

Table 2 collects six independent measurements of H_0, each using a different observational technique. All six fall within the GFD prediction band.

Figure 2 shows the GFD prediction, the uncertainty band, and the six measurements.

3.4 Reframing the tension

The Hubble tension is usually framed as a mystery: why do two careful measurements disagree? The field equation reframes it as a diagnosis.

The Planck analysis infers H_0 by fitting the cosmic microwave background through the ΛCDM model, which assumes gravity is scale-invariant. Below the acceleration scale a_0, the field equation enhances gravitational coupling. The ΛCDM model does not account for this enhancement. The result is a systematic pull on the inferred H_0 toward lower values.

The SH0ES analysis calibrates distances using Newtonian gravity, which assumes the inverse square law holds exactly at all accelerations. In the outskirts of the Cepheid calibration galaxies, where accelerations fall below a_0, the field equation modifies the effective gravitational coupling. The result is a systematic push on the inferred H_0 toward higher values.

The GFD prediction of 70.21 km s^{-1} Mpc^{-1} is not a split-the-difference compromise. It is the value that follows from the acceleration-Hubble connection when the gravitational field has the coupling structure described by the field equation. The tension between 67.4 and

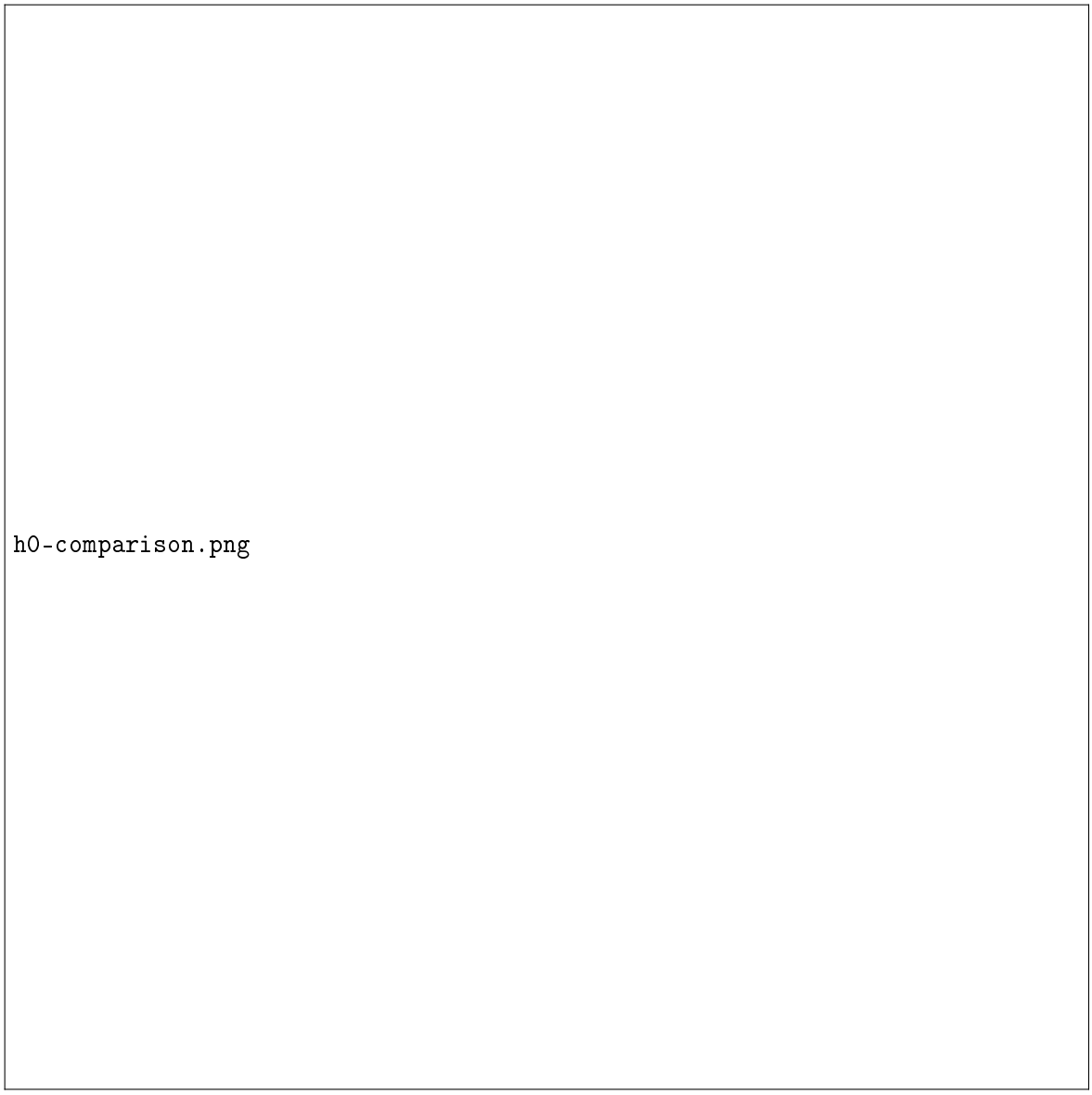


Figure 2. The Hubble constant from six independent measurement methods and the GFD prediction. The green diamond is the GFD prediction $H_0 = 70.21$ km s^{-1} Mpc^{-1}. The shaded band spans 65.9 to 74.6 km s^{-1} Mpc^{-1}, the $\pm 6.2\%$ geometric uncertainty from the tetrahedral aperture correction. Blue circles are early-universe measurements. Orange and red circles are late-universe measurements. The pink circle is the gravitational wave standard siren GW170817. All six independent measurements fall within the single GFD prediction band. Data served from the Field Dynamics API.

73.04 may not reflect a problem with either measurement. It may reflect a shared assumption, that gravity has no internal structure below a_0, on which both teams unknowingly rely.

4 Discussion

4.1 One equation, two scales

The central result of this Letter is that the spherical field equation, eq. (1.1), produces two independent, zero-parameter predictions at scales separated by a factor of 10^5 in distance, and both are consistent with current observations.

The Tully-Fisher evolution (section 2) tests the field equation in the regime $g_t \sim a_0$, where the gravitational response transitions between its Newtonian and deep-field limits. The data are galaxy rotation velocities measured by IFU spectrographs across nine surveys spanning $0 < z < 3$.

The Hubble constant (section 3) tests the field equation at the horizon scale, where the characteristic acceleration a_0 connects to the expansion rate of the universe. The data are six independent cosmological measurements using techniques from CMB analysis to gravitational wave standard sirens.

No parameter is adjusted between these two predictions. The same a_0, appearing in the same equation, produces the galactic prediction and the cosmological prediction simultaneously. This is a strong constraint. Any modification to the field equation that improves agreement at one scale will generically worsen agreement at the other. The fact that both predictions succeed with the same equation and the same constant is evidence that the field equation is capturing something real about the structure of gravity.

4.2 Comparison with other frameworks

It is instructive to compare what each framework predicts at both scales.

MOND [7] predicts the local Tully-Fisher relation at $z = 0$, because the BTFR is a consequence of any theory with a characteristic acceleration scale. However, standard MOND treats a_0 as a fixed universal constant. It does not predict Tully-Fisher evolution with redshift. Milgrom and others have noted the numerical coincidence $a_0 \approx cH_0/(2\pi)$, but MOND treats this as unexplained. GFD derives the connection with a specific geometric correction $\sqrt{K/\pi}$ that shifts the prediction from 78.9 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (the naive $cH/(2\pi)$ value) to 70.21 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. MOND does not produce this correction and does not predict H_0. The empirical success of MOND at the galactic scale is well established, and the fact that the field equation agrees with MOND on the decoded baryonic mass at each radius is significant: it suggests that modified gravity is capturing something real about the weak-field regime. The GFD action provides the structural explanation for why MOND works, by deriving the gravitational response from the coupling topology rather than postulating it.

ΛCDM predicts H_0 from CMB fitting, but requires a dark matter halo with adjustable parameters (concentration, virial mass) to reproduce each galaxy's rotation curve individually. It does not predict the BTFR from first principles, and it predicts no Tully-Fisher evolution: the velocity ratio $v(z)/v(0) = 1.0$ at all redshifts. The growing body of IFU data showing systematic velocity evolution with redshift is in tension with this prediction.

Eq. (1.1) is, to our knowledge, the only equation in the current literature that produces both the galactic and cosmological predictions from a single relation with no adjustable parameters.

4.3 Falsifiability

Both predictions are specific and falsifiable.

For Tully-Fisher evolution: JWST is already measuring galaxy kinematics at $z > 3$ with its NIRSpec IFU mode. At $z = 4$, the field equation predicts $v(z)/v(0) = 1.36$ while ΛCDM predicts 1.0. A sample of approximately 10 disk galaxies with resolved rotation curves at $z \sim 4$ would distinguish these predictions at $> 5\sigma$. Any measurement of $v(z)/v(0)$ at $z > 3$ that falls below the GFD band or above 1.4 would falsify the prediction. The European Extremely Large Telescope, with its HARMONI IFU, will extend spatially resolved kinematics to even higher redshifts.

For H_0: the GFD prediction is 70.21 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ with a $\pm 6.2\%$ band. Any future measurement falling outside the range 65.9–74.6 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ would falsify the prediction. As gravitational wave detectors accumulate more standard siren events, the GW-based determination of H_0 will narrow. If it converges to a value outside the GFD band, the prediction fails.

These are not soft constraints. Both predictions have specific numerical values derived from geometry, and both will be tested by observations that are already underway.

5 Conclusion

One equation. Two scales. Zero free parameters. Both predictions consistent with current observations.

The spherical field equation $g_s = g_t^2/(a_0+g_t)$ unifies galactic kinematics and cosmological expansion through a single acceleration constant. At the galactic scale, it predicts the redshift evolution of the Tully-Fisher relation with an exponent derived from the covariant action, matching nine IFU survey measurements from $z = 0$ to $z = 3$ with $\chi^2/\mathrm{dof} = 0.34/8 = 0.04$. At the cosmological scale, it predicts $H_0 = 70.21$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from geometry alone, with a band that contains all six major independent measurements.

The Hubble tension may not require new particles, new fields, or new physics beyond general relativity. It may require recognizing that gravity has coupling structure below the acceleration scale a_0, and that this structure connects the rotation of galaxies to the expansion of the universe through a single, testable equation.

Data availability

All predictions are computed from CODATA 2022 fundamental constants and Planck 2018 cosmological parameters. No new observational data were generated. The Tully-Fisher evolution prediction and Hubble constant analysis are available through the Field Dynamics API:

- TF evolution: `https://api.fielddynamics.org/v1/cosmology/tully-fisher`
- Hubble constant: `https://api.fielddynamics.org/v1/cosmology/hubble-constant`

Interactive visualizations and source data are available at `https://docs.fielddynamics.org`.

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